REVIEW

Optimizing Thrombolysis for Acute Ischemic Stroke

Adjunct Therapies and Multiple Dosing Strategies

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Abstract

Intravenous thrombolysis (IVT) and endovascular thrombectomy (EVT) are the keystones of current treatment for acute ischemic stroke. Most IVT candidates harbor medium or distal arterial occlusions, while EVT targets large artery occlusions. However, both treatments often fail to achieve sustained recanalization, microvascular reperfusion, and optimal clinical outcomes. Supplementing IVT with antithrombotic therapies has not shown consistent benefit, and alternative lytics such as tenecteplase provide, at best, a marginal improvement in outcome compared with alteplase. This review summarizes current evidence on multiple dosing strategies of IVT, including repeat administration of IVT in non-EVT candidates and IVT or intra-arterial thrombolytics in combination with EVT. We discuss the physiologic, pharmacokinetic, and clinical rationale for these approaches, highlight results from key clinical trials, and identify knowledge gaps to guide future research. Emerging data support novel fibrinolytic agents, sequential IVT dosing, and adjunct intra-arterial thrombolysis as promising strategies to enhance reperfusion and functional recovery while maintaining safety.

Introduction

Intravenous thrombolysis (IVT) and endovascular thrombectomy (EVT) are 2 complementary effective therapies that are the keystones of acute ischemic stroke (AIS) management. Both work primarily by dissolving or mechanically removing occlusive clots and restoring cerebral blood flow to brain tissue before it is irreversibly damaged. IVT is most effective for medium and distal arterial occlusions, which together are approximately 3 times more common than strokes due to large vessel occlusions (LVOs).[1] EVT very predictably recanalizes LVOs, although studies to date have shown a lack of benefit and possible increased risk in non-LVO lesions using current catheters.[2,3] Both IVT and EVT, either alone or in combination, will continue to be critical components of AIS therapy for the foreseeable future.

Both IVT and EVT have significant limitations. Initially, the concern was for bleeding complications, but with more timely treatment and better patient selection and management, the risk of bleeding is low. Symptomatic brain bleeding rates are approximately 2%–3% depending on the definition used.[4-6] The main limitation of both strategies is that roughly 50% of patients treated with either IVT or EVT remain functionally disabled at 90 days after stroke (defined as modified Rankin Scale [mRS] 2–6).[7] Among several factors that might lead to incomplete recovery, failure to recanalize the artery, recanalization followed by reocclusion, and sustained recanalization but failure to reperfuse the microcirculation have been identified as key culprits and possible targets for innovative approaches to enhance IVT and EVT. Improving IVT would be of particular utility because enhanced IVT could be administered widely and quickly in primary stroke centers and stroke-ready emergency departments (EDs) and in mobile stroke units.

The aim of this review was to examine the current literature on a new wave of adjunctive and multiple thrombolytic dosing strategies for both IVT alone and IVT combined with EVT. We explore the physiologic, pharmacokinetic, and clinical rationale behind these approaches,

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Glossary

DAPT = dual antiplatelet therapy; **DOAC** = direct oral anticoagulant; **ED** = emergency department; **END** = early neurologic deterioration; **EVT** = endovascular thrombectomy; **IAT** = intra-arterial thrombolysis; **ICAD** = intracranial atherosclerotic disease; **IMR** = incomplete microcirculatory reperfusion; **IVT** = intravenous thrombolysis; **LVO** = large vessel occlusion; **MeVO** = medium-size vessel occlusion; **mRS** = modified Rankin Scale; **TCD** = transcranial Doppler.

identify those most likely to be successful, and highlight knowledge gaps for future research. The Figure illustrates the major pathophysiologic mechanisms limiting recovery after reperfusion therapy and the therapeutic strategies discussed in this review.

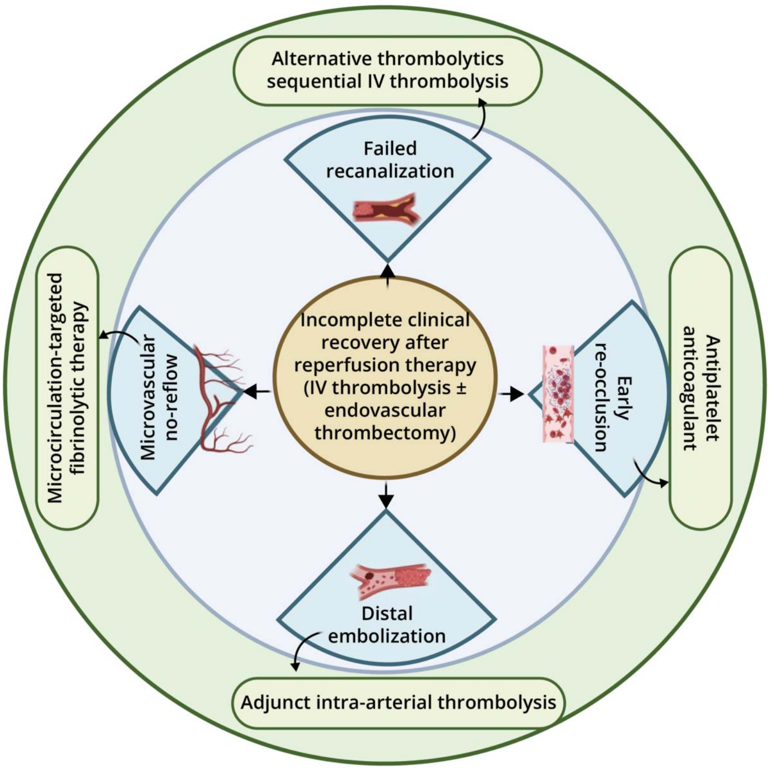
Single-Dose Intravenous Lytic Therapy Is Not Always Sufficient to Open the Artery

Studies have demonstrated that a single dose of IVT with either alteplase (tPA) or tenecteplase (TNK) may not achieve sustained or complete clot dissolution. In LVOs, the rates of successful immediate recanalization with intravenous tPA alone range from 8% to 32%,[8,9] with the highest rates observed with ultra-early treatment such as in mobile stroke units. Comparatively, TNK, a genetically modified thrombolytic agent with greater fibrin specificity and longer half-life, may offer modestly higher early recanalization rates.[10] When evaluating medium vessel occlusions (MeVOs), recanalization rates with tPA have been reported at 30% within 1 hour and up to 64% within 24 hours after treatment[11] and may be slightly higher with TNK.[7]

Recanalization rates after IVT and EVT are influenced by occlusion site, stroke severity, stroke mechanism, and the dynamics of thrombosis. Understanding these influences may help select patients for the adjunct therapies and multiple dosing strategies to be discussed further.

Acute recanalization with IVT is particularly poor in distal internal carotid artery and basilar artery occlusions.[12] Higher baseline NIHSS scores, reflecting greater thrombus burden and more proximal occlusion, are consistently associated with lower likelihood of IVT-induced recanalization.[13] Angiographic collateral grade further modulates treatment response, as better collateral status is associated with higher recanalization rates

Figure Pathophysiologic Mechanisms Limiting Clinical Recovery After Reperfusion Therapy and Potential Therapeutic Targets



Conceptual schematic illustrating mechanisms contributing to incomplete clinical recovery despite successful reperfusion. The center represents the clinical phenotype. The inner ring depicts key pathophysiologic mechanisms, including microvascular no-reflow and distal embolization. The outer ring highlights corresponding therapeutic strategies, including sequential intravenous thrombolysis, adjunct intra-arterial thrombolysis, and alternative thrombolytic approaches.

and improved clinical benefit, whereas poor collaterals may limit functional gains despite vessel reopening.[14] Advanced age has also been associated with lower rates of good clinical outcome even when angiographic reopening is achieved, suggesting that patient-related factors may modify the clinical impact of vessel recanalization.[15]

Beyond occlusion site and stroke severity, stroke mechanism further influences the durability and technical success of recanalization strategies. Patients with intracranial atherosclerotic disease (ICAD) have poorer outcomes after EVT compared with embolic occlusions, in part due to underlying fixed stenosis and high rates of acute reocclusion.[16] The same is likely also true in patients with a high burden of intracranial or extracranial atherosclerosis treated with IVT. Reocclusion, identified by serial transcranial Doppler (TCD) monitoring, occurred in 34% of patients with MCA occlusions who initially recanalized after IVT, and was associated with clinical deterioration after early improvement.[17] Tandem internal carotid artery occlusions, defined by concurrent cervical ICA disease and intracranial occlusion, are associated with lower response to IVT, more complex EVT procedures, and higher complication rates, including distal emboli and recurrent thrombosis.[18] Finally, cardioembolic clots pose their own complexities; they may be easier to remove or lyse but are prone to fragmentation, increasing the risk of distal embolization.[19,20] Together, these etiologic differences help explain variability in IVT and EVT recanalization success and clinical outcomes across stroke subtypes.

Finally, thrombus behavior is biologically dynamic, and clots may extend after patients receive IVT. Thrombus growth is associated with early neurologic deterioration (END), defined as a ≥ 4-point worsening on the 0–42 point NIH Stroke Scale (NIHSS). Among 120 patients, clot extension was present in 59% of those with unexplained END, compared with 29% without END.[21,22]

In addition to propagation, early reocclusion after initial lysis further limits the effectiveness of IVT. In a separate series of 84 patients treated with intravenous thrombolysis, reocclusion was observed in 17 patients (20%) after successful recanalization, with higher baseline NIHSS scores and carotid atherosclerotic disease identified as predictors.[23]

Impaired Microcirculatory Flow May Persist Even After Patients Receive IVT or EVT

Despite successful proximal recanalization after IVT or EVT, impaired microcirculatory perfusion often persists, commonly referred to as no-reflow or incomplete microcirculatory reperfusion (IMR). The original no-reflow description followed complete global interruption of cerebral blood flow in rabbits. Durations ≥5 minutes progressively increased the amount of tissue that failed to reperfuse despite restoration of proximal flow.[24] In rodent MCA occlusion models treated with tPA, IMR was attributed to endothelial swelling, capillary constriction, microthrombi, or embolic fragments clogging the microvascular network.[25] Clinical studies, primarily in patients treated with EVT, have confirmed that angiographic recanalization does not always equate to completely restored issue reperfusion. IMR is clinically assessed using perfusion imaging, such as perfusion CT or perfusion-weighted MRI, by detecting persistent tissue hypoperfusion (e.g., reductions in relative cerebral blood flow or volume) despite angiographic recanalization. These perfusion deficits allow identification of microvascular no-reflow at the tissue level. In a systematic review and meta-analysis of 13 studies including 719 patients, persistent IMR after successful macrovascular recanalization occurred in 29% (95% CI 21%–37%) and was associated with reduced rates of functional independence at 90 days.[26] IMR has also been associated with larger final infarct volume.[27] More recent clinical imaging data confirm that persistent microcirculatory perfusion deficits can be detected after successful macrovascular reperfusion and are associated with early neurologic impairment.[28]

Distal embolization of small clot fragments dislodged downstream is another contributor to mismatch between arterial reopening and tissue reperfusion. In patients undergoing EVT, distal emboli were detected on postprocedural MRI in 21%–28%[29] of patients with anterior circulation stroke. IVT may also paradoxically increase embolic fragmentation during clot lysis, especially with fibrin-rich, partially organized thrombi.[30] Intravenous tPA administered before EVT resulted in 25% incidence of distal embolization vs 7% without tPA.[31]

Together, these studies illustrate that IVT and EVT alone often do not achieve complete sustained recanalization, and even when proximal arterial flow is restored, they do not consistently reestablish adequate microvascular perfusion, resulting in larger infarct size and worse clinical outcome.

Combining IVT With Antithrombotic Therapy

Efforts to augment the efficacy of single-dose IVT have mainly focused on combining tPA or TNK with antiplatelet or anticoagulant agents. While aiming to mitigate early reocclusion, enhance microvascular perfusion, and improve clinical outcomes, these strategies carry the risk of increased hemorrhagic complications.

Antiplatelet Strategies

Current guidelines recommend withholding antiplatelet therapy for 24 hours after IVT because of concerns for hemorrhagic complications. A trial combining IVT with aspirin alone found no clear functional benefit and raised safety concerns.[32] Similarly, intravenous aspirin (300 mg) given within 90 minutes of tPA produced more sICH (4.3% vs 1.6%

with tPA alone), with no observed benefit in functional outcome.[33] Dual antiplatelet therapy (DAPT), most commonly aspirin plus clopidogrel, has demonstrated benefit compared with IVT in patients with acute minor nondisabling ischemic stroke, with a lower risk of symptomatic intracranial hemorrhage.[34] However, there is no evidence supporting the safety or efficacy of simultaneous DAPT and IVT (e.g., READAPT cohort study [pmc.ncbi.nlm.nih.gov/articles/PMC11963945]) (Table 1).

Glenzocimab, a humanized monoclonal antibody fragment directed against platelet glycoprotein VI, has been studied in combination with standard of care IVT ± EVT. An early-phase clinical study suggested a favorable safety profile and reductions in intracranial hemorrhage and mortality when added to standard thrombolysis ± EVT,[35] but a subsequent phase II/III study (ACTISAVE; NCT05070260) showed no significant improvement in the primary or secondary efficacy end points.

The glycoprotein IIb/IIIa inhibitor eptifibatide has been safely administered during or immediately after low-dose or full-dose IVT in phase II studies.[36] Eptifibatide has also been evaluated as an adjunct to EVT without a clear signal of excess hemorrhage.[37] A pivotal multicenter randomized controlled trial compared the safety and efficacy of IVT alone with IVT combined with adjunctive intravenous eptifibatide or the direct thrombin inhibitor argatroban started by the end of the 1-hour tPA infusion. The trial was stopped early for futility because neither combination improved functional outcomes.[5] The study also included patients with LVO who underwent EVT after IVT; there was no benefit of eptifibatide in that subset either.

A recently reported large controlled trial from China evaluated tirofiban, a peptidomimetic that inhibits glycoprotein IIb/IIIa function when administered intravenously. Tirofiban significantly improved excellent functional outcomes in patients with AIS when started within 60 minutes after IVT compared with IVT plus placebo; the nondisabled outcome at 90 days was 65.9% vs 54.9%. However, sICH occurred in 1.7% of tirofiban-treated patients vs 0% in those with placebo.[38,39] The dose of tirofiban in this study was proportionately low compared with the cardiac dose, and most secondary outcomes were not positive. Tirofiban has also been tested as an alternative to IVT in patients undergoing EVT, demonstrating safety but no superiority for tirofiban.[40] Taken together, these data suggest that further evaluation of tirofiban in combination with IVT may be warranted. Logically such studies might target patients with a substantial burden of atherosclerosis, given the propensity of such patients to experience reocclusion.

Anticoagulant Strategies

Combining IVT with anticoagulants has yielded mixed or unfavorable results. Post-IVT administration of heparin demonstrated modest reductions in recurrent ischemic events that were negated by significantly elevated risks of sICH and has largely been abandoned in modern practice.[41]

Phase II studies suggested that adjunctive administration of argatroban, a direct thrombin inhibitor, given concurrently with full-dose tPA may be safe and potentially enhance early recanalization.[42,43] However, a pivotal phase III trial did not confirm these results. Mortality was in fact higher with argatroban, although the argatroban arm was underpowered because enrollment into that arm was terminated early by the trial's adaptive design.[5,44]

The effect of direct oral anticoagulants (DOACs) on recanalization and outcomes in patients receiving IVT is unknown. Because of the ubiquitous use of DOACs in patients with atrial fibrillation, IVT is often considered in patients with cardioembolic stroke who have recently taken one of these medications. Owing to concerns about bleeding risk, European Stroke Organization and American Heart Association

Table 1 Summary of Clinical Trials Evaluating Early Antiplatelet Therapy After Intravenous Thrombolysis in Acute Ischemic Stroke

Study	Thrombolytic	Antiplatelet regimen	Timing of DAPT/ASA	Primary outcomes	Key findings
Toni et al. (1998)[e18]	rt-PA	Aspirin (300 mg IV)	Within 90 min after IVT	Functional outcome at 3 mo, sICH	Increased sICH; no improvement in functional outcomes
Zinkstok et al. (2010)[e19]	rt-PA	Aspirin (300 mg IV)	Within 90 min after IVT	Functional outcome at 3 mo, sICH	Increased sICH; no improvement in functional outcomes
Zhao et al. (2021)[e20]	rt-PA	Aspirin + clopidogrel	24 h after IVT	Functional outcome at 90 d, sICH	No increase in sICH; favorable functional outcomes
Zeng et al. (2024)[e21]	rt-PA	Aspirin + clopidogrel	24 h after IVT	Functional outcome at 90 d, sICH	Improved prognosis; no increase in hemorrhage or mortality
Hamayal et al. (2025)[e22]	rt-PA	Aspirin + cslopidogrel	Within 24 h after IVT	Functional outcome at 90 d, sICH	Better functional outcomes; reduced risk of stroke recurrence and sICH

Abbreviations: ASA = aspirin; DAPT = dual antiplatelet therapy; IVT = intravenous thrombolysis; rt-PA = recombinant tissue plasminogen activator; sICH = symptomatic intracranial hemorrhage.

guidelines have recommended against IVT use within 48 hours after last DOAC intake when specific coagulation tests are unavailable.[45] However, based on a systematic review showing no significant increase in sICH or mortality among patients who received IVT within 48 hours of DOAC exposure compared with IVT in non–DOAC-taking controls,[46] DOAC use has been downgraded to a relative contraindication to IVT in recent American Heart Association guidelines.[47] A randomized trial is underway comparing IVT with no IVT in stroke patients on DOACs and will evaluate both safety and effect on clinical outcome. If this study demonstrates the safety of combining IVT with DOAC use, a subsequent step would be to examine whether DOAC exposure results in higher recanalization rates and better outcomes after IVT than IVT alone.

Alternative Thrombolytic Agents

Efforts to improve IVT outcomes have led to the investigation of thrombolytics beyond tPA. These include TNK, desmoteplase, urokinase, pro-urokinase, and reteplase.

TNK is a modified form of human tissue plasminogen activator that binds to fibrin and converts plasminogen to plasmin. In vitro studies demonstrated that in the presence of fibrin, TNK conversion of plasminogen to plasmin is increased relative to its conversion in the absence of fibrin.[48] This fibrin specificity decreases systemic activation of plasminogen, and the resulting degradation of circulating fibrinogen, compared with a molecule lacking this property, could potentially decrease the incidence of bleeding.[49]

Pooled and meta-analyses have largely confirmed a slight superiority of TNK in early recanalization rates and noninferiority in functional outcomes compared with tPA.[50-53] A large Canadian registry–linked randomized study (n = 1,600) found that 0.25 mg/kg TNK was noninferior to tPA regarding functional outcomes and had comparable safety.[54] Based largely on these noninferiority results, the FDA approved the use of TNK for AIS and recommended weight-based tiered dosing used in that trial.

Desmoteplase, tested in patients with perfusion mismatch up to 9 hours after stroke onset, showed encouraging results in phase II trials but failed in phase III studies.[55] Urokinase and pro-urokinase were investigated primarily through intra-arterial administration and showed improved recanalization of MCA occlusions, but high hemorrhage rates. Limited data with intravenous use prevented further adoption.[56,57] In the TRUST (ThRombolysis of Urokinase for minor Stroke Trial), intravenous urokinase did not improve functional outcomes compared with standard medical therapy although it was safe.[58]

Double bolus reteplase (18 mg × 2, 30 minutes apart) was compared with tPA in 1,412 patients from China with AIS treated within 4.5 hours of symptom onset. Reteplase resulted in higher probability of excellent functional recovery at 90 days, without increased risk of sICH, although there was a modestly higher rate of overall intracranial bleeding and adverse events.[59]

Beyond these established agents, newer approaches targeting fibrinolysis enhancement are under investigation, including von Willebrand factor inhibition and alpha-2-antiplasmin (α2-AP) antagonism. Strategy for Improving Stroke Treatment Response (SISTER)[e1] is a phase II trial of an α2-AP monoclonal antibody, and Recanalization in Acute Ischemic Stroke Patients (RAISE)[e2] is a phase II trial of an oligonucleotide that inhibits von Willebrand factor. Both studies aim to improve reperfusion and reduce infarct growth in patients with a persisting ischemic penumbra beyond the time window for administration of currently approved IVT.

Taken together, studies to date have demonstrated, at best, marginal superiority of TNK compared with tPA in clinical outcomes. TNK is rapidly replacing alteplase because of its convenient administration and probable superior rates of recanalization. Research is ongoing to identify alternative or complementary thrombolytic strategies. Exactly how they would fit into the existing armamentarium of IVT and EVT will be an important component of their evaluation. A summary of alternative intravenous thrombolytic agents and major clinical trial findings is provided in Table 2.

Higher Single Doses of Intravenous Lytic

Multiple studies have demonstrated a dose-dependent increase in hemorrhagic risk with both tPA and TNK, suggesting that higher single doses of existing IVT will not be safe.

For tPA, pilot studies tested doses ranging from 0.6 mg/kg to 1.2 mg/kg. Doses exceeding 0.9 mg/kg were associated with unacceptably high rates of sICH, ranging from 10% to 20% in some subgroups.[e3,e4] Based on these findings, the pivotal NINDS tPA stroke trial selected the now-standard 0.9 mg/kg dose administered within 3 hours of symptom onset and reported a sICH rate of 6.4%.[e5]

For TNK, early dose-finding studies also revealed that bleeding risk increases with higher doses. An Australian trial compared 0.1, 0.25, and 0.4 mg/kg TNK. The 0.4 mg/kg arm was terminated early because of a higher rate of sICH (15%)[e4,e6] These findings were supported by a large, pragmatic, multicenter randomized trial of 0.4 mg/kg TNK across a broad stroke population with a high proportion of mild strokes. While overall sICH rates were similar between the TNK and tPA arms (3%), post hoc analysis showed that among patients with moderate-to-severe strokes, TNK 0.4 mg/kg produced higher mortality and poorer functional outcomes.[e7]

Table 2 Summary of Alternative Intravenous Thrombolytic Agents Studied in Acute Ischemic Stroke

Author	Agent	Outcome
Palaiodimou et al.[50]	Tenecteplase (meta-analysis)	TNK was associated with a statistically significant increase in excellent functional outcome without significant differences in sICH or mortality
Hacke et al.[55]	Desmoteplase (DIAS-2)	No clinical benefit on primary composite outcome at 90 d; low sICH; mortality similar to placebo
Tao et al.[58]	Urokinase/pro-urokinase (TRUST trial)	Did not improve excellent functional outcome at 90 d compared with best medical therapy; no significant differences in sICH or mortality
Li et al.[59]	Reteplase	More likely to result in an excellent functional outcome than alteplase; no significant increase in sICH

Abbreviations: DIAS = Desmoteplase in acute ischemic stroke; sICH = symptomatic intracranial hemorrhage; TNK = Tenecteplase.

Multiple Sequential Doses of IVT

Biologically, the concept of administering a second IVT dose hinges on the presence of a residual thrombus substrate. Binding of plasminogen activators to fibrin receptors in the clot is instantaneous after an IV bolus but dissipates over the next 30–60 minutes. As already discussed, despite an initial dose of lytic, many patients experience only partial or transient recanalization because of high clot burden, reocclusion, or incomplete thrombus dissolution. These residual thrombi remain susceptible to enzymatic degradation by plasminogen activators if sufficient plasminogen substrate is available, which supports the plausibility of a second thrombolytic dose exerting meaningful biological activity.

In most disease states, more than a single dose of a medication is required to eliminate the underlying pathology. However, until recently, only single dosing of IVT has been attempted. As mentioned previously, double bolus reteplase resulted in better outcome without substantial increased risk compared with standard-dose tPA.[58]

A comparative case-controlled study conducted at 2 stroke centers in France evaluated a dual-IVT approach for patients with AIS due to MeVO. All patients received initial standard-dose tPA, and one center administered TNK 0.25 mg/kg approximately 2 hours after the tPA bolus in patients who did not exhibit early recanalization on MRI/MRA. Compared with matched patients at a second center receiving standard-dose tPA alone, patients receiving the dual-IVT strategy showed higher rates of 24-hour recanalization (82% vs 65%) and better 90-day outcome (nondisabled outcome 69% vs 44%). In dual-IVT patients, sICH occurred in only 2 and 1 had a psoas hematoma requiring transfusion vs 2 with sICH in the tPA group.[11]TNK has advantages for this dual-dosing approach because of its higher fibrin specificity compared with tPA, ease of single-bolus administration, and pharmacokinetics that suggest that 2 full doses might be safe.[47] Assuming the first TNK dose is given within the first 3 hours after stroke onset, the second dose could be given approximately 60 minutes later, still within the approved 4.5-hour time window for IVT. The maximal lytic effect of TNK occurs approximately 30 minutes after the bolus.[e8] The agent has a plasma half-life of approximately 22 minutes, allowing for near-complete systemic clearance within 60–70 minutes (approximately 2.7 half-lives).[e9] At that point, only approximately 15% of the original dose remains, pharmacologically equivalent to 0.038 mg/kg. Administering a second full dose of 0.25 mg/kg at this time would result in a cumulative exposure of approximately 0.288 mg/kg, which remains within or below the safety margins previously explored in dose-finding trials. A pilot safety study of this approach in patients qualifying for IVT but not EVT and who do not initially respond to the first dose based on persisting neurologic deficit is underway[e10] (NCT06801054).

Intra-Arterial Thrombolytics as an Adjunct to Endovascular Therapy

EVT With Intra-Arterial Thrombolytics (Without Previous IVT)

In patients who are ineligible for IVT because of contraindications or delayed presentation, the primary strategy is often EVT without preceding IVT. In such cases, adjunctive intra-arterial thrombolysis (IAT) has been studied to target residual distal emboli, incomplete microvascular reperfusion, or persistent occlusion. A recent meta-analysis of randomized controlled trials demonstrated that adding IAT after successful EVT significantly improved excellent functional outcomes (mRS 0–1 at 90 days), without increasing rates of symptomatic intracranial hemorrhage or mortality.[e11] These data suggest that IAT may offer a safe and effective adjunctive option in EVT settings.

EVT With Intra-Arterial Thrombolysis and IVT

The most intensive lytic combination under current evaluation is the triple therapy approach, where patients first received IVT followed by EVT, and then IAT if reperfusion remains incomplete. The Chemical Optimization of Cerebral Embolectomy (CHOICE) trial demonstrated that intra-arterial tPA administration after successful EVT significantly improved functional independence rates without compromising safety. Half of the patients had also received IVT with standard-dose tPA before EVT, and there was no interaction between pre-EVT IVT and risk. This study specifically targeted patients with suboptimal reperfusion after EVT, addressing the clinical challenge of IMR after technically successful recanalization.[e12]

EVT With Intra-Arterial Thrombolytics (With or Without IVT)

A recent meta-analysis pooling 6 trials (n = 1,971) compared EVT alone with EVT combined with IAT, regardless of whether IVT was used. The analysis showed that IAT after EVT significantly increased excellent functional outcomes (mRS 0–1), with no increase in symptomatic intracranial hemorrhage, mortality, or systemic bleeding.[e13]

Taken together, these studies suggest that IAT represents a safe and promising therapeutic avenue for enhancing microvascular reperfusion and functional recovery after EVT, with or without pre-EVT IVT. While the IAT lytic dose in these trials was one-fourth to one-half of the standard IVT dose, it is likely that drug levels at the clot surface with these IAT doses might approximate those achieved after full IVT dosing. Further studies to identify optimal IAT dosing are needed. Key studies evaluating adjunct intra-arterial thrombolysis after EVT are summarized in Table 3.

Who Are the Best Candidates for Multiple Dosing of IVT or IAT?

The studies of administering multiple doses of IVT or EVT + IAT have used imaging-based identification of residual occlusion or incomplete reperfusion to guide thrombolytic redosing. Beyond confirming vessel patency, advanced diagnostic tools can further refine therapeutic decision making. CTA, MRI, and MRA not only localize thrombus but also characterize clot burden and collateral status, factors known to influence both the likelihood and durability of recanalization. Complementing static imaging, continuous TCD monitoring provides real-time assessment of early recanalization, reocclusion, and microembolic activity after IVT. After EVT, perfusion imaging with CT or MRI can detect persistent tissue-level hypoperfusion despite macrovascular reopening, identifying patients with no-reflow who may benefit from adjunctive intra-arterial therapy.[e14] Together, these modalities offer a framework for individualized escalation strategies based on both vascular and tissue-level information.

However, among IVT patients who are not candidates for EVT, repeat vascular imaging is not feasible at many stroke centers because of limited access to rapid MRA, and dye and radiation exposure with CTA. Moreover, reliance solely on imaging may exclude a substantial subset of patients with persistent disabling symptoms and undetected distal vessel occlusions or IMR. Sixty-nine percent of patients who are otherwise eligible for IVT but not EVT do not have visible occlusion on initial vascular imaging, suggesting the presence of distal occlusions that are radiographically occult.[e15] At least 50% of these patients had disabled outcome with standard single-dose IVT.

Another strategy would be to guide eligibility for a second dose of IVT using clinical criteria, specifically the NIHSS, rather than repeat imaging. Early recanalization correlates with rapid clinical improvement, while a lack of improvement or secondary worsening in NIHSS scores predicts persistent occlusion and poor outcome.[e16,e17] Thus, patients whose NIHSS scores remain elevated or worsen after an initial dose of IVT may represent ideal candidates for redosing, if repeat noncontrast brain CT excludes bleeding complications from the initial dose. A delayed, second IVT dose strategy tailored to clinical response offers a personalized and physiologically sound approach. It allows the initial pharmacologic effect of IVT to be assessed before escalating therapy, while remaining within approved treatment windows and respecting drug clearance kinetics.

In the context of adjunct IAT after EVT, the question remains whether all patients should receive IAT or whether its use should be reserved for a selected subset. Unlike IVT-only pathways, in patients undergoing EVT, clinicians often have more precise information of reperfusion status through angiographic imaging, enabling a more targeted approach. Clinically, persistent neurologic deficits despite apparent complete clot retrieval may also identify patients with ongoing perfusion deficits that are not apparent on standard imaging.

Table 3 Studies on Adjunct Intra-Arterial Thrombolysis After EVT

Study	IV thrombolysis before EVT	IA agent (dose/notes)	Key outcome (mRS 0–1 at 90 d)	Symptomatic ICH/mortality
Jiang X et al.[e23] (meta-analysis)	Variable IVT use	Various IAT (alteplase/urokinase/tenecteplase, variable)	↑ Excellent outcome with adjunct IAT (RR ∼1.25, $p < 0.01$)	No significant increase in sICH ($p > 0.05$) or mortality ($p > 0.05$)
CHOICE, Chamorro et al.[e13]	Full-dose IV alteplase	Alteplase, low-dose IA (0.225 mg/kg, max 22.5 mg)	59% vs 40.4% excellent outcome ($p = 0.047$)	sICH 0% vs 3.8% ($p = 0.30$); mortality 8% vs 15% ($p = 0.17$)
PEARL, Yang et al.[e24]	Full-dose IV alteplase	Alteplase, low-dose IA (0.225 mg/kg)	44.8% vs 30.2% excellent outcome ($p = 0.01$)	sICH 4.3% vs 5.0% ($p = 0.84$); mortality 17.1% vs 11.3% ($p = 0.26$)

Abbreviations: EVT = endovascular thrombectomy; IA = intra-arterial; IAT iIntra-arterial thrombolysis; ICH = intracranial hemorrhage; mRS = modified Rankin Scale; RR = relative risk; sICH = symptomatic intracranial hemorrhage.

Therefore, adjunct IAT after EVT may be most appropriate for those with either imaging evidence of residual distal occlusion or clinical evidence of ongoing ischemia, rather than being administered universally.

Conclusion

Despite advances in reperfusion therapies, many patients with AIS continue to experience suboptimal outcomes because of incomplete vessel recanalization and tissue reperfusion. Emerging evidence supports further study of antiplatelet therapy immediately after IVT, novel fibrinolytics, sequential dosing regimens of IVT, and adjunct IAT after EVT. These strategies have the potential to improve vessel recanalization, enhance microvascular reperfusion, and translate into better functional recovery. Future clinical trials should focus on personalized, clinically guided redosing protocols that leverage imaging and neurologic criteria to maximize benefit and minimize risks.

Author Contributions

M. Shahrivari: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design. J.C. Grotta: drafting/revision of the manuscript for content, including medical writing for content.

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